MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11746908 LOW VOLTAGE POWER SUPPLY CONTROL LOGIC

This specification is approved for use by the US Army Armament Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly: 11746908 Low Voltage Power Supply Control Logic which is the A2A2 component of the Electronic Unit, 11743131.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment: Acquisition, Maintenance and Disposition of
MIL-STD-45662	Calibration System Requirements

FSC 1240

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

DRAWINGS

ARRADCOM

11746908	Circuit Card Assembly (Low Voltage Power Supply Control Logic)

Inspection Equipment

11750221	Test Set - Low Voltage Power Supply

Packaging Data Sheet

P11746908	Packaging of Circuit Card Assembly: 11746908 (Low Voltage Power Supply Control Logic)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The low voltage power supply control logic circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11746908 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (See 4.5.1)

3.1.1 Function. The assembly shall provide the following functions:

	Designation
a.	Continuity
b.	Output voltage characteristics
c.	15 Volt sense and overcurrent
d.	Q-switch drive
e.	Sync signals A and B
f.	Q-switch drive regulator

3.1.2. General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	73° + 18°F
b.	Relative humidity	50 percent ± 30 percent
c.	Atmospheric pressure	28.5 + 2.0 -3.0 in. Hg.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

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3.3.1 Continuity. With no power or signals applied, direct continuity, resistance equal to or less than 2.0 ohms, shall exist between pins J1-20 and J1-30. (See 4.6.2.1)

3.3.2 Output voltage characteristics. With the input power conditions established as specified in table II, the output voltages shall be as specified in table II. (See 4.6.2.2)

TABLE I. Loads, power and signals.

Item	Conditions	Characteristics	Connections
1.0	Loads		Connected as follows:
1.1	Resistors	390 ohms + 5 percent 1/4 watt.	One each between pin J1-30 and each of pins J1-22 and J1-24.
1.2	Resistors	10 kilohms +5 percent 1/4 watt.	One each between pin J1-30 and each of pins J1-36, J1-37 and J1-39.
1.3	Resistors	150 ohms +5 percent 3 watts in series with Item 2.2	One to each of pins J1-3, J1-4, J1-5 and J1-6.
1.4	Resistors	150 ohms +5 percent 1/2 watt.	Between pins J1-21 and J1-30.
1.5	Resistor	120 ohms +5 percent 1/2 watt.	Between pins J1-23 and J1-30.
1.6	Resistor	15.4 kilohms +1.0 percent 1/4 watt.	Between pins J1-1 and J1-7, when specified.
1.7	Resistor	22.0 ohms +1.0 percent 10 watts in series with Item 2.2.	To pins J1-3 and J1-4 when specified.

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TABLE I. Loads, power and signals. - Continued

Item	Conditions	Characteristics	Connections
1.8	Resistor	1 kilohm ±5 percent 2 watts in series with Item 2.3	To pin J1-10.
1.9	Resistor	1 kilohm ±5 percent 1/4 watt.	Between pins J1-29 and J1-30.
1.10	Resistor	240 ohms ±5 percent 5 watts in series with Item 2.3	To pin J1-8.
1.11	Resistor	1 kilohm ±5 percent 1/4 watt.	Between pins J1-38 and J1-30.
1.12	Resistor	270 ohms ±5 percent 5 watts.	Between pins J1-13 and J1-20.
1.13	Resistor	300 kilohms ±5 percent 1/4 watt.	Between pins J1-19 and J1-30
2.0	Power Sources		Unless otherwise specified, applied between pins:
2.1	5.0 Vdc.	Tolerance: ± 0.2 Vdc Ripple: 25 mV p-p max.	J1-26 (+) and J1-30 (-), and as specified
2.2	14.0 Vdc	Tolerance: ±0.5 Vdc Ripple: 25 mV p-p max.	Applied as specified between pins: J1-40 (+) J1-28 (+) and J1-30 (-).
2.3	Tank Power	Adjustable: 0 to 45 Vdc Ripple: 200 mV p-p max.	Applied as specified.

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TABLE I. Loads, power and signals. - Continued

Item	Conditions	Characteristics	Connections
2.4	3.0 Vdc	Tolerance: ±0.2 Vdc Ripple: 25 mV p-p max.	J1-14 (+) and J1-15 (-)
2.5	10.0 Vdc	Tolerance: ±0.5 Vdc Ripple: 25 mV p-p max.	Applied as specified, between pins J1-40 (+) and J1-30 (-).
3.0	Signal Source		
3.1	$\overline{\text{FIRE}}$	Logic one: 4.0 ±1.0 V Source impedance: 2 kilohms ±5 percent. Logic zero: 0.2 ±0.2 V	Applied as specified between pins J1-2 and J1-30.

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TABLE II. Output voltage characteristics.

	Input power conditions 4/ 6/		DC output at pins of J1 3/, 5/					
Item	J1-31 2/	J1-40 1/	36	37	38	39	29	19
1	28 ± 2V	OC	+11.2+2V-1V	+11.2+2V-1V	1V max.	+12+2V-1V	+13 ± 1V	NA
2	41.5 ± 0.5V	14V	NA	NA	+12+2V-1V	+12+2V-1V	+0.2 +0.2V	6.9 +0.7V
3	36.5 ± 0.5V	14V	NA	NA	NA	NA	+13 ± 1V	NA
4	13.0 ± 0.5V	14V	NA	NA	NA	NA	+13 ± 1V	NA
5	10.0 ± 0.5V	14V	NA	NA	NA	NA	+0.2 ± 0.2V	NA
6	28 ± 2V	10V	NA	NA	NA	+9 ± 1V	NA	NA

1/ Input power per items 2.2 and 2.5 of table I.
2/ Input power per item 2.3 of table I.
3/ Referenced to J1-30.
4/ OC = Open circuit
5/ NA = Not applicable.
6/ Pin J1-28 is 14 volts for all items. See note 1/.

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3.3.3 15 volt sense and overcurrent. With the input power, item 2.2 of table I, and the 240 ohm load, item 1.10 of table I, applied as specified therein and J1-9 connected to J1-30, the output voltages shall be as specified in table III and figure 1. (See 4.6.2.3)

3.3.4 Q-switch drive. With the input power, item 2.2 of table I,

and the FIRE, item 3.1 of table I, applied as specified and the input power item 2.3 of table I, applied at 28 ±2 Vdc between pins J1-31 (+) and J1-30 (-), the output pulse train between J1-22, J2-24 and J1-30 (rtn) shall be as specified on figure 2. With the input power, item 2.3 of table I, applied at 41.5 ±0.5 Vdc, the output signals between pins J1-22, J1-24 and J1-30 (rtn) shall be 1 +1 Vdc. (See 4.6.2.4)

3.3.5 Sync signals A and B. With the input power, item 2.2 of table I, applied as specified, the output signals between pins J1-21, J1-23, and J1-30 (rtn) shall be as specified in figure 3. (See 4.6.2.5)

3.3.6 Q-switch drive regulator. With the input power, item 2.4 of table I, applied as specified, J1-15 connected to J1-16 and the input power, item 2.3 of table I, applied at 31 ±1 Vdc between pins J1-12 (+) and J1-20 (-), the output voltage between pins J1-13 and J1-20 (rtn) shall be 26.4 ±3.0 Vdc. With the interconnection removed from J1-15 and J1-16 and +5 Vdc, item 2.1 of table I, applied between pins J1-14 (+) and J1-20 (-), the output voltage between J1-13 and J1-20 (rtn) shall be 1 +1 Vdc. (See 4.6.2.6)

3.4 Environmental.

3.4.1 Shock. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ± 4 gravity units (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the profile of figure 1. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)

3.4.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -25°F to +140°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

3.4.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to +160°F. (See 4.6.3.3, 4.6.3 4 and 4.6.3.5)

3.4.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65° to +160°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

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TABLE III. 15V sense and overcurrent.

Item	Input power conditions 3/ 9/	Loads		Output voltages at pins of J1 3/ 6/				
	J1-7 2/	J1-1 4/ 5/	J1-3, -4 6/ 7/	8	3	4	5	6
1	16.8 +0.5V	OC	NA	1 +1V	1 +1V	1 +1V	14 +1V	14 +1V
2	13.6 +0.5V	OC	NA	1 +1V	Fig. 1	Fig. 1	Fig. 1	Fig. 1
3	13.6 +0.5V	15.4 kilohms	NA	1 +1V	NA	NA	NA	14 +1V
4	13.6 +0.5V	OC	22.0 ohms	13.6 +0.5V	NA	NA	NA	14 +1V 8/

1/ Input power per item 2.2 of table I.
2/ Input power per item 2.3 of table I.
3/ Referenced to J1-30.
4/ OC = Open circuit.
5/ Load per item 1.6 of table I.
6/ NA = Not applicable.
7/ Load per item 1.7 of table I, applied after input power conditions are established.
8/ During overcurrent condition.
9/ Pin J1-40 is 14 volts for all items. See note 1/.

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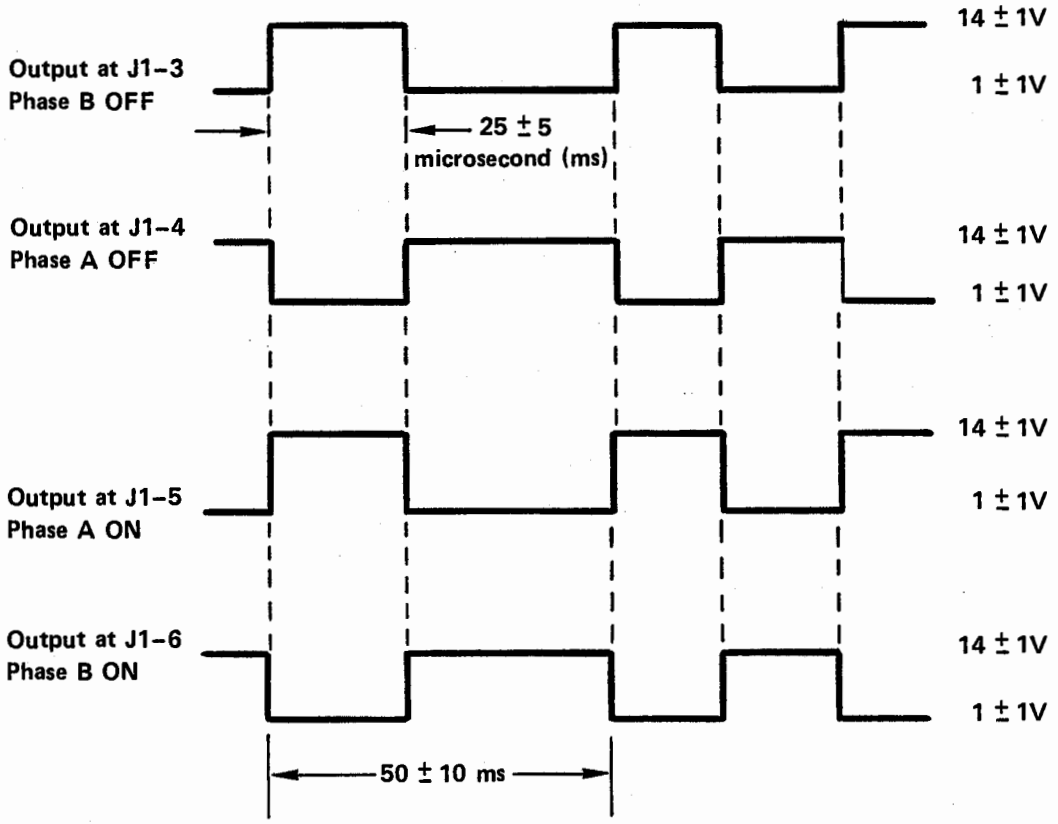


FIGURE 1. Phase A and B waveforms.

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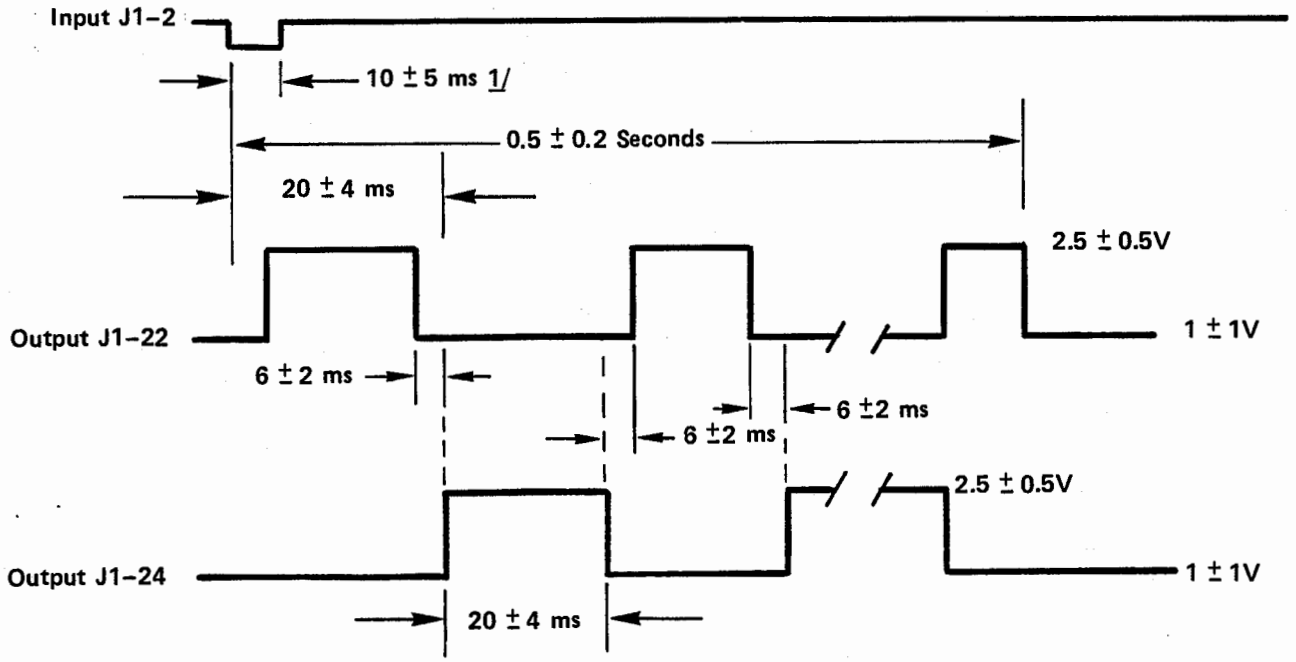


FIGURE 2. Q-switch drive waveforms.

1/ Pulse widths and time intervals are specified at 50% of peak pulse amplitude.

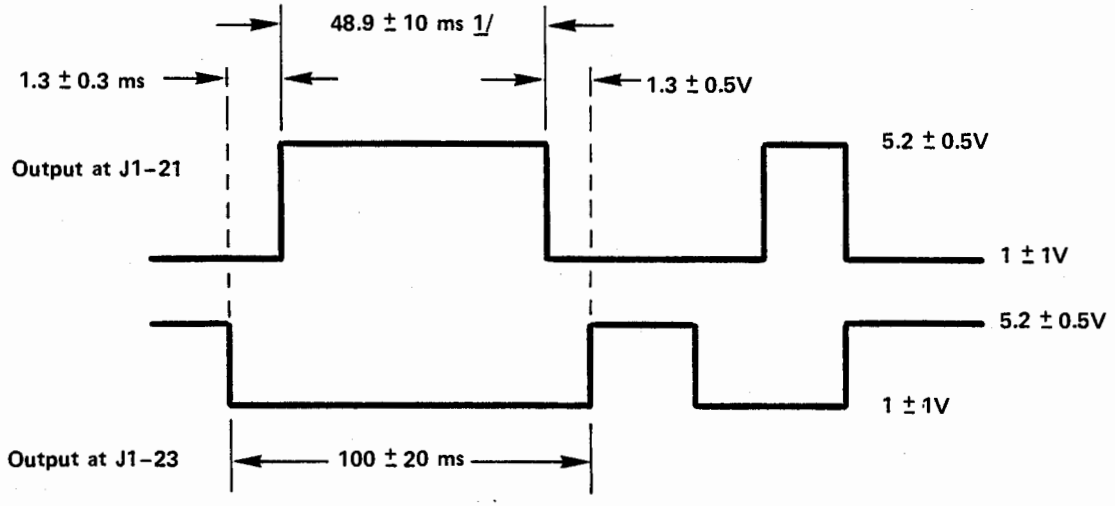


FIGURE 3. Sync signal A and B.

1/ Pulse widths and time intervals are specified at 50% of peak pulse amplitude.

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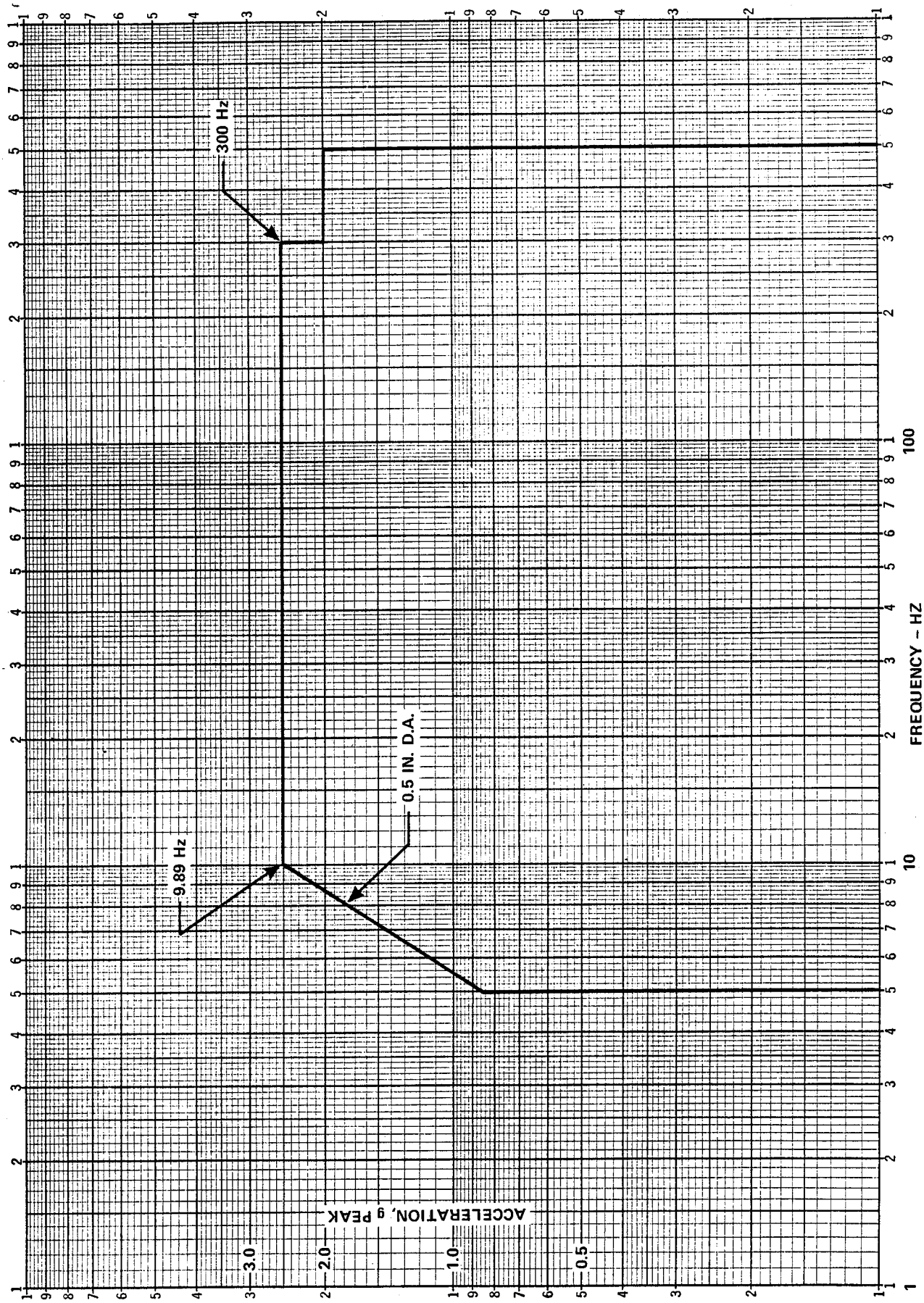


FIGURE 4. Vibration profile.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first fifteen production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in tables V, VI and VII.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

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TABLE V. Performance tests.

Item	Characteristic	Requirement	Test Procedure
101	Continuity	3.3.1	4.6.2.1
102	Output voltage characteristics	3.3.2	4.6.2.2
103	15 volt sense and overcurrent	3.3.3	4.6.2.3
104	Q-switch drive	3.3.4	4.6.2.4
105	Sync signals A and B	3.3.5	4.6.2.5
106	Q-switch drive regulator	3.3.6	4.6.2.6

TABLE VI. General tests.

Item	Characteristic	Requirement	Test Procedure
301	Fabrication	3.1	Applicable drawings - Visual
302	General specifications	3.1.2	MIL-F-13926 - Visual

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TABLE VII. Environmental tests.

Item	Characteristic	Requirement	Test Procedure
303	Shock	3.4.1	4.6.3.1
304	Vibration	3.4.2	4.6.3.2
305	High Temperature	3.4.3 and 3.4.4	4.6.3.4
306	Low Temperature	3.4.3 and 3.4.4	4.6.3.4
307	Temperature cycling	3.4.3 and 3.4.4	4.6.3.5

4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A and C
b. Component assembly of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.6.2.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in table VI.

4.5.2.2 Environmental sample and tests. Three assemblies, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table VII shall be applied separately to each assembly in the sample.

4.5.2.3 Acceptance. Where any one assembly of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subject to the provisions of MIL-STD-105.

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4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table VI and only the test of item 307 in table VII shall be applied.

4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in table V and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11746908.

4.6 Methods of inspection.

4.6.1 Test equipment. Unless the assembly is procured apart from the next higher assembly, the test equipment in table VIII shall be used where specified to perform the required test.

TABLE VIII. Test equipment.

Item	Part Number	Nomenclature
1	11750221	Test Set - Low Voltage Power Supply

4.6.2 Performance tests.

4.6.2.1 Continuity. Use item 1 of table VIII or suitable equipment to determine compliance with 3.3.1.

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4.6.2.2 Output voltage characteristics. Use item 1 of table VIII or suitable equipment to determine compliance with 3.3.2.

4.6.2.3 15 volt sense and overcurrent. Use item 1 of table VIII or suitable equipment to determine compliance with 3.3.3.

4.6.2.4 Q-switch drive. Use item 1 of table VIII or suitable equipment to determine compliance with 3.3.4.

4.6.2.5 Sync signals A and B. Use item 1 of table VIII or suitable equipment to determine compliance with 3.3.5.

4.6.2.6 Q-switch drive regulator. Use item 1 of table VIII or suitable equipment to determine compliance with 3.3.6.

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure I, figue 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 ±4 g with a time duration of 18 ±3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table V.

4.6.3.1.2 High intensity. Mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 100.0 ±10.0 g with a time duration of 1.5 ±0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table V.

4.6.3.2 Vibration. Vibration A as specified in 4.6.3.2.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.3.2.2 shall be conducted only for quality conformance inspection.

4.6.3.2.1 Vibration A. Mount the assembly in a suitable vibration fixture and subject the assembly to the vibration test specified in MIL-STD-810, method 514.2 procedure VIII, except that the test level shall be the vibration curve shown in figure 4 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of table V.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

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4.6.3.3 High temperature. Subject the assembly to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 140°F. The performance tests of table V shall be applied.

4.6.3.4 Low temperature. Subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of table V shall be applied.

4.6.3.5 Temperature cycling. With power and loads of table I applied, subject the assembly to the temperature profile specified in figure 5. At the conclusion of this test, subject the assembly to the performance tests of table V.

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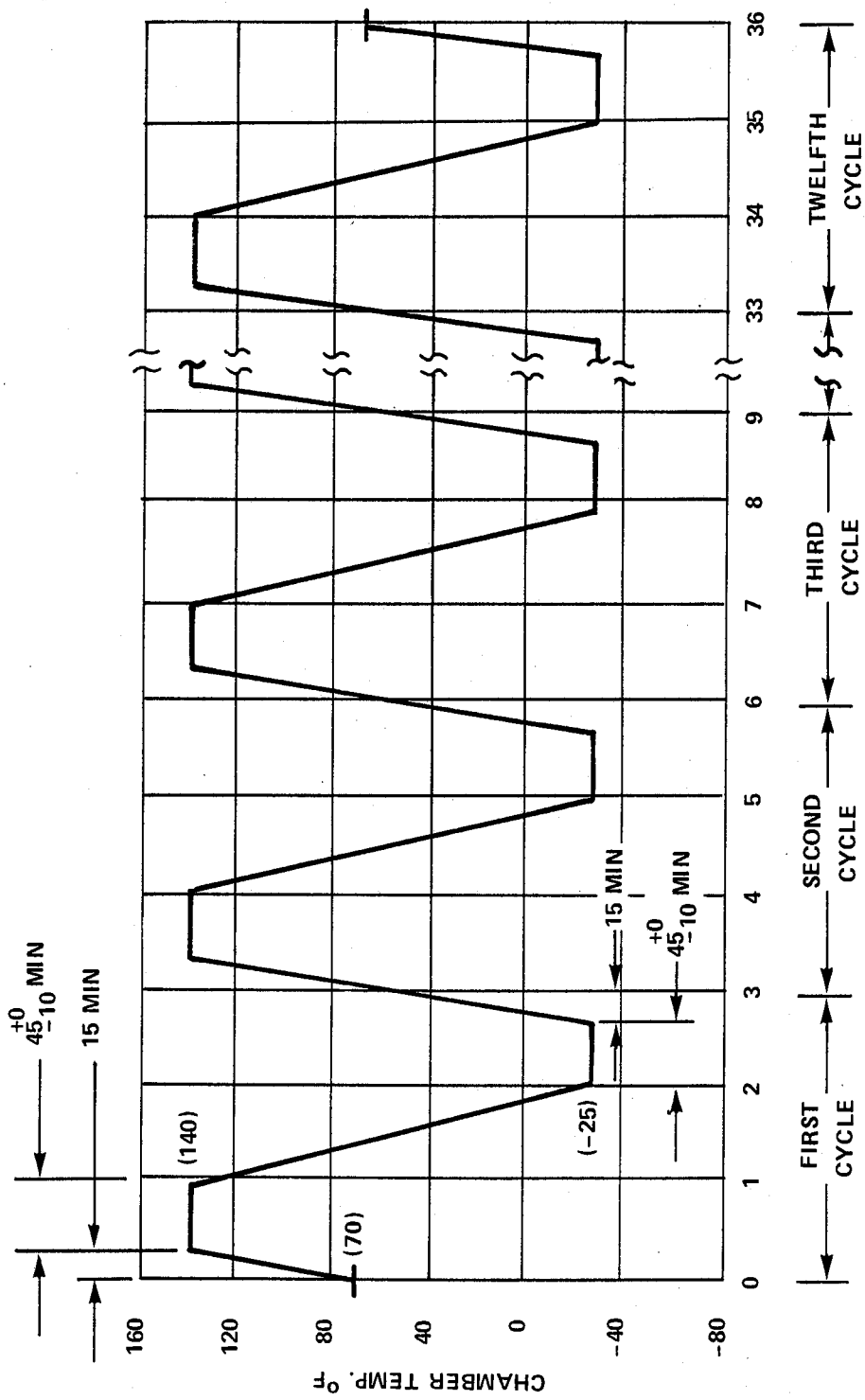


FIGURE 5. Temperature cycling.

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5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11746908. The level of protection shall be as specified in the procurement document. (See 4.5.6)

6. NOTES

6.1 Intended use. The Low Voltage Power Supply Control Logic Circuit Card Assembly is a functional assembly of the Low Voltage Power Supply, intended for use in the Laser Fire Control Rangefinder AN/VVG-2. The major functions of the assembly are to provide:

a. Regulator drive waveforms
b. 10 kHz square wave sync signals
c. Over current sensing
d. Over voltage sensing
e. Relay drive

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Special tests and examination as required, the type of inspection required and the activity responsible for testing and its facility.

c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. First article inspection requirements.

6.3 Previous revisions of MIL-C-48610B was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

Custodian:
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Preparing Activity:
Army-AR
Project 1230-A799

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